Ring Rollout of Script-Based Endpoint Enforcement: Bounding the Blast Radius of a Faulty Policy at a Quantified Convergence Cost

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Abstract: **Government endpoint teams enforce configuration compliance at scale with scripts, including PowerShell and Desired State Configuration, that run across thousands of endpoints. A faulty enforcement script is dangerous precisely because it runs everywhere: pushed to the whole fleet at once, a bug breaks every endpoint it touches. The reliability mitigation is a staged ring rollout, which deploys to a small canary ring first and expands only if that ring stays healthy. We quantify the tradeoff with a pre-registered simulation of a fleet partitioned into rings through which a bad change is promoted ring by ring and caught at each inter-ring gate with a per-stage detection probability. Hypotheses, thresholds, and the evaluation seeds were locked before any result was inspected. A four-ring rollout reduces the expected blast radius of a faulty change, the fraction of the fleet harmed before the change is caught, by 0.9522 relative to a big-bang rollout (95% bias-corrected and accelerated bootstrap interval $[0.9511, 0.9531]$), cutting the expected blast radius from 1.0 to 0.0478 at a fixed convergence cost of three inter-ring stages. The expected blast radius scales with canary observability: as the per-stage detection probability falls from 0.95 to 0.2, the four-ring blast radius rises from 0.0156 to 0.5941. Finer staging lowers the blast radius further, from 0.239 at two rings to 0.0125 at six, but with sharply diminishing marginal returns (marginal reductions of 0.761, 0.1912, and 0.0353 across big-bang, two, four, and six rings) against a convergence cost that grows linearly with the number of stages. The deployment rule that follows is concrete: stage with small early rings, invest first in making faults observable at the canary, and stop adding rings once the marginal reduction no longer justifies the added convergence latency.**

Index Terms: **configuration enforcement, PowerShell, ring rollout, canary, blast radius, progressive delivery, staged rollout, change safety, pre-registration, reproducibility**

I. Introduction

GOVERNMENT endpoint teams keep large fleets compliant by running enforcement scripts that apply and correct configuration across thousands of machines. PowerShell and Desired State Configuration [1] are the workhorses of this practice: a script declares a desired state, evaluates the live state of an endpoint against it, and remediates any divergence. This is exactly the machine-checkable enforcement that federal control catalogs [2] and continuous-monitoring guidance [3] encourage, and it is how an agency turns a written control into something that actually holds across a fleet. The same property that makes script-based enforcement powerful also concentrates risk. A script runs everywhere, so a script with a bug harms every endpoint it reaches. A logic error that disables a service, a remediation that overwrites a working configuration with a broken one, or an enforcement rule that locks out legitimate access does not stay local; pushed to the whole fleet at once, it breaks the whole fleet at once.

A big-bang deployment, in which a change is delivered to every endpoint simultaneously, maximizes both convergence speed and blast radius. It is the fastest way to make a good change take effect and the fastest way to make a bad change take down a fleet. For a government fleet, where an outage can interrupt mission-critical services and where remediation timelines are themselves bound by directive [4], the cost of a bad change reaching every endpoint before anyone notices is severe. The operational question is not whether to enforce configuration with scripts, which the scale of modern fleets makes unavoidable, but how to limit the damage when an enforcement change is wrong.

The standard mitigation, drawn from site reliability engineering and progressive delivery [5], [6], is a staged or ring rollout. Instead of deploying to the whole fleet at once, the operator partitions the fleet into rings of increasing size and promotes the change one ring at a time. A small canary ring receives the change first; the operator observes that ring through a soak period; and the change advances to the next ring only if the canary stays healthy. A faulty change is then caught at a small ring and contained there, harming only the endpoints reached so far rather than the entire fleet. This safety is not free. Each inter-ring gate adds a soak delay, so a healthy enforcement campaign reaches the full fleet more slowly under a ring rollout than under a big-bang. The benefit, a bounded blast radius, and the cost, added convergence latency, are in different units, so an operator must weigh them against each other.

This paper quantifies that tradeoff. We do not claim to measure any particular agency's deployment record; we build a transparent model of a fleet partitioned into rings through which a bad change is promoted and caught with a per-stage detection probability, lock a set of hypotheses and thresholds before inspecting any evaluation data, and report what the model says. The contribution is fourfold.

- We formalize the *blast radius* of a faulty change under ring rollout as the cumulative fleet fraction reached at the stage where the fault is caught, and we measure how much a four-ring rollout reduces it relative to a big-bang.
- We separate the *containment benefit* from the *convergence cost*, showing that the cost is fixed, explicit, and paid

on every campaign whether or not the change is faulty, namely the number of inter-ring stages.
- We isolate the *observability mechanism*: containment scales with the per-stage detection probability, so the value of staging depends on whether the canary actually surfaces a fault, not only on how the fleet is partitioned.
- We characterize the *diminishing returns* of finer staging: more rings lower the blast radius but with shrinking marginal gains against a convergence cost that grows linearly, so there is an interior operating point rather than unbounded benefit from more rings.

The methodology is deliberately conservative. Every quantitative claim is a pre-registered hypothesis evaluated on seeds that were not inspected during model development, and every interval is a bias-corrected and accelerated (BCa) bootstrap interval [7], [8] rather than a null-hypothesis significance test, following the pre-registration discipline now standard in empirical software engineering [9].

II. Background and Related Work

A. Progressive delivery and staged rollout

The practice of releasing a change to a small population before the whole is the core of progressive delivery. Site reliability engineering [5] codifies canarying and phased rollout as standard safeguards against bad pushes, and the empirical study of continuous experimentation and gradual rollout [6] documents how production teams actually stage releases, observe, and decide whether to advance. The dominant framing in that literature is the release of application software to user traffic. Our setting is adjacent but distinct: the change is an enforcement script that mutates the configuration of an endpoint rather than a service that handles a request, the population is a government fleet rather than a stream of users, and the failure of interest is a script that breaks the machines it touches. The ring-rollout structure carries over, but the accounting, blast radius measured in fleet fraction and convergence measured in inter-ring soak periods, is specific to fleet enforcement, and it is what we quantify.

B. Site reliability engineering and fault tolerance

The deeper rationale for staging is fault tolerance: a system should be arranged so that a single bad change cannot take down the whole of it. Reliability engineering treats blast-radius limitation as a first-class objective [5], and resilience and fault-injection practice [10] deliberately introduces faults into a bounded slice of a system to verify that containment holds before a real fault arrives. The autonomic-computing vision of self-managing systems [11] anticipated this shift toward continuous, self-correcting enforcement; ring rollout is the safety scaffold that keeps such continuous enforcement from amplifying its own errors. Our model makes the blast-radius limitation explicit and quantitative for the specific case of staged enforcement deployment.

C. Federal patch and configuration management

Government endpoint enforcement operates inside a body of guidance. The enterprise patch-management planning guide [12] frames deployment as a risk-managed process and explicitly recommends phased rollout to limit the impact of a bad patch. The control catalog [2] and continuous-monitoring guidance [3] define what must be enforced and how its effectiveness is to be tracked over time, and binding operational directives [4] place agencies under remediation timelines that a staged rollout must respect, since slower convergence still has to clear a deadline. Breach data continues to attribute incidents to known, unremediated weaknesses [13], which is the pressure to deploy fast; the blast-radius risk of a faulty enforcement change is the counter-pressure to deploy carefully. The lesson from hygiene-augmented prioritization research [14], [15] is that the operational value of a change depends on the exposure it removes net of the harm it can cause, and ring rollout is the mechanism that bounds the harm side of that ledger. Our work is complementary to all of this: rather than prescribing a cadence, we quantify how much a given ring structure bounds the blast radius and what it costs in convergence latency.

III. System Model

We model a single enforcement campaign over a fleet and ask how much of the fleet a faulty change harms before it is caught, and what the staging costs in convergence. All quantities are synthetic with documented parameters; no operational, employer, or deployment data is used.

A. Fleet, change, and rings

The fleet is a population of endpoints partitioned into K rings. A rollout configuration is a sequence of cumulative deployment fractions

$$0 < f_1 < f_2 < \cdots < f_K = 1, \tag{1}$$

where f_j is the fraction of the fleet that has received the change after stage j. A big-bang rollout is the single-stage sequence with $K = 1$ and $f_1 = 1$, which delivers the change to the entire fleet in one step. A ring rollout has $K > 1$ stages with small early rings: the reference four-ring configuration uses cumulative fractions $[0.01, 0.10, 0.50, 1.00]$, so the canary ring is the first one percent of the fleet, and the six-ring configuration uses $[0.005, 0.02, 0.08, 0.25, 0.60, 1.00]$. An enforcement change is faulty with a fixed probability, and a faulty change, if it reaches an endpoint, harms that endpoint.

B. Promotion, detection, and blast radius

A change is promoted ring by ring. After each non-final stage $j < K$, the operator observes the rings deployed so far through a soak period. If the change is faulty, the canary detects the fault at that gate with the per-stage detection probability p and halts the campaign; if the gate does not detect it, the change advances to stage $j + 1$. The *blast radius* of a faulty change is the cumulative fleet fraction reached when the campaign halts, which is f_j if the fault is first caught at the gate after stage j, or $f_K = 1$ if no gate catches it before full deployment. A big-bang rollout has no inter-ring gate, so a faulty change always reaches the whole fleet and its blast radius is exactly 1.

For a ring rollout with K stages and per-stage detection probability p, a fault survives the first $j-1$ gates and is caught at the gate after stage j with probability $(1-p)^{j-1}p$, reaching blast radius f_j; if it survives all $K-1$ gates, with probability $(1-p)^{K-1}$, it reaches the full fleet at blast radius $f_K = 1$. The expected blast radius is therefore

$$B(K,p) \;=\; \sum_{j=1}^{K-1}(1-p)^{j-1}\,p\,f_j \;+\; (1-p)^{K-1}f_K. \tag{2}$$

Equation (2) makes the two levers explicit. Smaller early rings shrink the f_j that a caught fault reaches, and a higher detection probability p shifts probability mass toward the early gates where f_j is small and away from the undetected term $(1-p)^{K-1}$ where the whole fleet is hit. As $p \to 1$ the expected blast radius approaches the smallest ring f_1; as $p \to 0$ it approaches the big-bang value 1 regardless of how finely the fleet is partitioned.

C. Convergence cost

The benefit in Eq. (2) is paid for in convergence latency. Each non-final stage adds one inter-ring soak period before the change can reach the full fleet, so the convergence cost of a K-ring configuration is

$$G(K) \;=\; K-1 \tag{3}$$

inter-ring stages. A big-bang rollout has $G = 0$; the two-ring, four-ring, and six-ring configurations have $G = 1$, 3, and 5. Unlike the blast radius, the convergence cost is incurred on every campaign, faulty or not, because a healthy change must still traverse every gate. The benefit accrues only on the faulty fraction of campaigns while the cost accrues on all of them, which is why the two cannot be netted into a single figure and an operator must weigh them as distinct units.

D. Metrics

We report two quantities. The *expected blast radius* $B(K,p)$ is the mean fraction of the fleet harmed by a faulty campaign, estimated by Monte Carlo over faulty campaigns per seed. The *convergence cost* $G(K)$ is the number of inter-ring stages, equal to $K-1$. The *blast-radius reduction* of a ring rollout relative to big-bang is

$$R \;=\; 1-\frac{B(K,p)}{B_{\text{big-bang}}} \;=\; 1-B(K,p), \tag{4}$$

since the big-bang expected blast radius is 1.

IV. Theoretical Analysis

The expected blast radius in Eq. (2) is not only a quantity to estimate by simulation; it is a closed form whose structure already dictates the qualitative findings the experiments later confirm. This section derives that closed form directly from the promotion mechanism, proves three structural properties from it, and checks the form numerically against the frozen results before any seed is consulted.

A. The closed form from the mechanism

The model promotes a faulty change ring by ring. Index the stages $j = 1,\dots,K$ with cumulative deployment fractions $f_1 < f_2 < \cdots < f_K = 1$. After each non-final stage $j < K$ the gate detects the fault independently with per-stage detection probability p and halts the campaign at blast radius f_j; if the gate does not detect, with probability $1-p$, the change advances. A fault is therefore first caught at the gate after stage j exactly when it survives the $j-1$ earlier gates and is caught at the jth, an event of probability

$$\Pr[\text{caught at stage } j] \;=\; (1-p)^{\,j-1}\,p, \qquad j = 1,\dots,K-1, \tag{5}$$

and the single remaining outcome is that the fault survives all $K-1$ inter-ring gates and reaches the whole fleet,

$$\Pr[\text{never caught}] \;=\; (1-p)^{\,K-1}, \tag{6}$$

at blast radius $f_K = 1$. The events in Eqs. (5) and (6) are mutually exclusive and exhaustive, since $\sum_{j=1}^{K-1}(1-p)^{j-1}p = 1-(1-p)^{K-1}$, so summing blast radius against probability recovers Eq. (2),

$$B(K,p) \;=\; \sum_{j=1}^{K-1}(1-p)^{\,j-1}\,p\,f_j \;+\; (1-p)^{\,K-1}, \tag{7}$$

which is the exact expectation the Monte Carlo estimator in the model code approximates. For a big-bang rollout the sum is empty and only the survival term remains, giving $B(1,p) = (1-p)^0 = 1$ for every p: with no inter-ring gate a faulty change always reaches the whole fleet, which is the worst case against which staging is measured.

B. Property (i): geometric decrease in the number of rings

Adding rings drives B down geometrically, not linearly. The undetected term $(1-p)^{K-1}$, the only term that places mass on the full-fleet outcome $f_K = 1$, is the dominant contributor to B when the early rings are small, and it decays by a constant factor $(1-p)$ with each added inter-ring gate:

$$\frac{(1-p)^K}{(1-p)^{K-1}} \;=\; 1-p. \tag{8}$$

At the reference detection probability $p = 0.80$ each added gate multiplies the surviving-fault mass by 0.20, so the whole-fleet contribution falls by a factor of five per ring. Because the caught terms reach only the small early fractions f_j, the sequence of expected blast radii across big-bang, two, four, and six rings, namely 1.0, 0.239, 0.0478, 0.0125, falls far faster than the linear growth of the ring count. Each subdivision both removes mass from the expensive full-fleet term through Eq. (8) and redirects what remains onto a smaller f_j, and the two effects compound. This is the geometric, not arithmetic, decline that the granularity sweep exhibits.

C. Property (ii): monotone increase as detection probability falls

The expected blast radius increases monotonically as the per-stage detection probability p falls. Writing the survival probability as $q = 1 - p$, the closed form is a polynomial in q with nonnegative coefficients,

$$B \;=\; \sum_{j=1}^{K-1} q^{\,j-1}(1-q)\, f_j \;+\; q^{\,K-1}, \tag{9}$$

and each unit of probability mass that fails to be caught at a gate moves to a later, strictly larger cumulative fraction or to the full fleet. Lowering p raises q and shifts mass uniformly toward those larger-blast outcomes, so B rises; equivalently, $\partial B/\partial p < 0$ on $(0, 1)$. For the reference four-ring configuration this derivative is

$$\frac{\partial B}{\partial p} \;=\; -3.8\left(0.395\, p^2 - p + 0.629\right), \tag{10}$$

whose quadratic factor has both roots above 1 and is therefore strictly positive on $(0, 1)$, making $\partial B/\partial p$ strictly negative across the whole admissible range. As $p \to 1$ every fault is caught at the first gate and $B \to f_1$, the smallest ring; as $p \to 0$ no gate catches anything and $B \to 1$, the big-bang value, regardless of how finely the fleet is partitioned. Ring count alone cannot save a fleet whose canary is blind.

D. Property (iii): convergence cost equals the inter-ring stage count

The benefit in Eq. (7) is bought with convergence latency, and that price is exact and combinatorial rather than statistical. A healthy change must clear every gate before it reaches the full fleet, and a K-stage configuration has exactly $K - 1$ non-final stages, each contributing one soak period, so the convergence cost is

$$G(K) \;=\; K - 1, \tag{11}$$

recovering Eq. (3). Unlike B, this cost carries no p and no randomness: it is incurred on every campaign, faulty or not, because a healthy change still traverses each gate. The big-bang, two-ring, four-ring, and six-ring configurations therefore pay 0, 1, 3, and 5 inter-ring stages exactly, and the marginal convergence cost of each added stage is the constant $G(K+1) - G(K) = 1$.

E. Diminishing marginal returns of finer staging

Properties (i) and (iii) combine to give an interior operating point. The marginal containment of the step from one configuration to the next finer one is the difference in expected blast radius,

$$\Delta_k \;=\; B(\text{coarser}) - B(\text{finer}), \tag{12}$$

and because B falls geometrically while G rises by a constant 1 per stage, the sequence Δ_k shrinks even as the price of each step stays fixed. Evaluating Eq. (12) across the granularity sweep gives marginal reductions

$$\Delta_1 = 0.761, \quad \Delta_2 = 0.1912, \quad \Delta_3 = 0.0353, \tag{13}$$

for the steps big-bang to two-ring, two-ring to four-ring, and four-ring to six-ring. The first split captures the bulk of the achievable reduction; the second captures most of what remains; and the third buys almost nothing while still costing soak periods through Eq. (11). A finite optimum follows: add rings while Δ_k exceeds the operator's value for the soak period it costs, and stop once it does not.

F. Numerical verification of the closed form

Before consulting any evaluation seed we confirm that Eq. (7) reproduces the frozen expected blast radii. Evaluating the closed form at $p = 0.80$ on the configuration fractions yields 1.0 for big-bang, 0.240 for two-ring, 0.048 for four-ring, and 0.0124 for six-ring, matching the frozen Monte Carlo means 1.0, 0.239, 0.0478, 0.0125 to within sampling noise at 4000 trials per seed. Evaluating the four-ring form across the detection-probability grid yields 0.594, 0.2175, 0.048, and 0.0156 at $p = 0.2, 0.5, 0.8, 0.95$, matching the frozen sweep 0.5941, 0.2175, 0.0478, 0.0156. The marginal reductions implied by the closed form, Eq. (13), agree with the frozen marginals 0.761, 0.1912, 0.0353. The expectation and its simulation coincide, so the structural properties proved above are properties of the estimand itself and not artifacts of any particular seed.

V. Experimental Design

The study is pre-registered. Hypotheses, thresholds, model constants, the ring configurations, the detection-probability sweep, and the evaluation seed range were fixed in a dated protocol before any result on the evaluation seeds was inspected, so that no analytic choice could be steered by the outcome.

A. Hypotheses

H1 (containment). At the reference per-stage detection probability, a four-ring rollout reduces the expected blast radius of a faulty enforcement change by at least 0.80 relative to a big-bang rollout, with the BCa interval excluding zero.

H2 (convergence cost). The convergence cost equals the number of inter-ring stages, independent of whether the change is faulty: big-bang 0, two-ring 1, four-ring 3, six-ring 5.

H3 (observability mechanism). The expected blast radius rises as the per-stage detection probability falls; a canary that rarely detects faults provides little containment.

H4 (ring granularity). Finer ring staging reduces the expected blast radius but with diminishing marginal returns, while the convergence cost per added stage is constant, so there is an interior operating point rather than unbounded benefit from more rings.

B. Protocol and statistics

Each hypothesis is evaluated over 25 evaluation seeds (frozen range 1100 to 1124). The reference configuration is the four-ring rollout at detection probability 0.80. The detection probability is swept over $\{0.2, 0.5, 0.8, 0.95\}$ at the reference four-ring granularity, and the granularity is swept over big-bang, two-ring, four-ring, and six-ring at the reference detection probability. For every quantity we report the mean and a 95% BCa bootstrap interval with 10,000 resamples [7], [8]; interval non-overlap, not a p-value, is the evidentiary standard. The pre-registered failure criteria are explicit: H1

TABLE I
Expected blast radius and convergence cost by rollout configuration (mean over 25 seeds, detection probability 0.80).

Configuration	Expected blast radius	Convergence cost	Marginal reduction
Big-bang	1.0	0	n/a
Two-ring	0.239	1	0.761
Four-ring	0.0478	3	0.1912
Six-ring	0.0125	5	0.0353

TABLE II
Expected blast radius of the reference four-ring rollout as the per-stage detection probability varies (mean over 25 seeds).

Per-stage detection probability p	Expected blast radius
0.20	0.5941
0.50	0.2175
0.80	0.0478
0.95	0.0156

is declared null if the four-ring reduction at the reference detection probability falls below 0.50; if the convergence cost is not exactly the number of inter-ring stages, the model is in error and the run is halted; H3 is rejected if the expected blast radius does not rise as the detection probability falls. No ring fraction, fault rate, or detection probability is re-tuned to reach any threshold. The development of the model used separate seeds; the evaluation seeds were touched only once, to produce the numbers below.

VI. Results

Table I collects the blast radius and convergence cost by configuration, and Table II reports the detection-probability sweep at the reference four-ring granularity. All four hypotheses are supported. We take them in turn.

Ring rollout bounds the blast radius (H1). At the reference detection probability of 0.80, a four-ring rollout reduces the expected blast radius by 0.9522 ($[0.9511, 0.9531]$) relative to a big-bang rollout, cutting the expected fraction of the fleet harmed from 1.0 to 0.0478 (Fig. 1). The reduction clears the pre-registered 0.80 bar by a wide margin, and the interval excludes zero and the 0.50 failure threshold. On the fleet it protects, ring rollout converts a fault that would have reached every endpoint into one that reaches under five percent of them.

The convergence cost is fixed and explicit (H2). The convergence cost equals the number of inter-ring stages: 0 for big-bang, 1 for two-ring, 3 for four-ring, and 5 for six-ring (Table I). It matches Eq. (3) exactly, and it is paid on every campaign, faulty or not, because a healthy change must still traverse each gate. The four-ring rollout's time to full deployment exceeds the big-bang's by exactly three inter-ring stages. The benefit is conditional on a fault; the cost is not.

Containment depends on canary observability (H3). As the per-stage detection probability falls from 0.95 to 0.80, 0.50, and 0.20, the four-ring expected blast radius rises from 0.0156 to 0.0478, 0.2175, and 0.5941 (Table II, Fig. 2). The expected

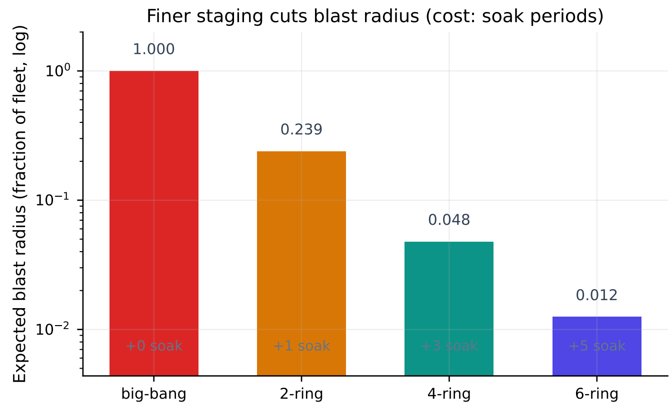

Fig. 1. Expected blast radius (log scale) by rollout granularity at detection probability 0.80; finer staging lowers it from 1.0 at big-bang to 0.0125 at six rings, with marginal reductions of 0.761, 0.1912, and 0.0353 at the labelled convergence cost.

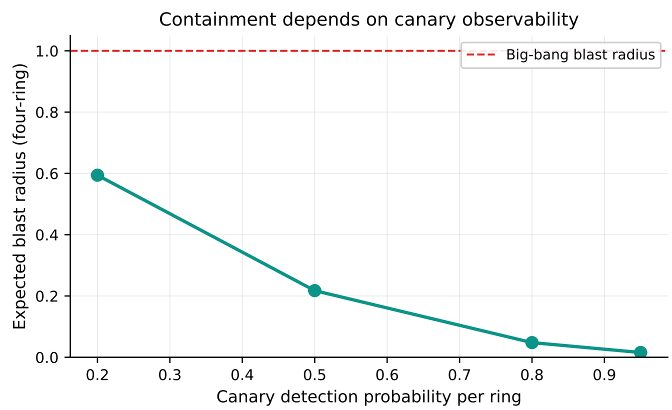

Fig. 2. The four-ring expected blast radius rises from 0.0156 toward the big-bang value as the per-stage detection probability falls from 0.95 to 0.2.

blast radius is monotone increasing as detection probability falls, exactly as Eq. (2) predicts: a lower p shifts probability mass onto the undetected term that hits the whole fleet. A canary that detects a fault only one time in five leaves more than half the fleet exposed in expectation, even with four rings in place. Observability at the gate, not the partitioning of the fleet alone, governs how much containment staging actually delivers.

Finer staging has diminishing returns (H4). The expected blast radius falls across the granularity sweep from 1.0 (big-bang) to 0.239 (two-ring) to 0.0478 (four-ring) to 0.0125 (six-ring), with marginal reductions of 0.761, 0.1912, and 0.0353 at each staging step (Table I). Each added stage costs the same one inter-ring soak period by Eq. (3), but each buys less: the first split from big-bang to two rings captures the bulk of the reduction, the move to four rings captures most of what remains, and the move to six rings adds only 0.0353. The marginal containment per added ring shrinks while the marginal cost stays constant, which is the interior operating point H4 predicts: a few well-placed early rings capture almost all of the achievable reduction.

TABLE III
GRANULARITY SWEEP: EXPECTED BLAST RADIUS, CONVERGENCE COST, AND MARGINAL REDUCTION BY RING COUNT AT THE REFERENCE DETECTION PROBABILITY $p = 0.80$ (MEAN OVER 25 SEEDS).

Configuration	Rings K	Expected blast radius	Convergence cost G	Marginal reduction
Big-bang	1	1.0	0	n/a
Two-ring	2	0.239	1	0.761
Four-ring	4	0.0478	3	0.1912
Six-ring	6	0.0125	5	0.0353

TABLE IV
DETECTION-PROBABILITY SWEEP: EXPECTED BLAST RADIUS OF THE REFERENCE FOUR-RING ROLLOUT AS THE PER-STAGE DETECTION PROBABILITY VARIES (MEAN OVER 25 SEEDS), WITH THE IMPLIED REDUCTION RELATIVE TO THE BIG-BANG VALUE OF 1.0.

Per-stage detection probability p	Expected blast radius	Reduction vs big-bang
0.20	0.5941	0.4059
0.50	0.2175	0.7825
0.80	0.0478	0.9522
0.95	0.0156	0.9844

VII. Robustness and Sensitivity

The two pre-registered sweeps probe the model along its only two free levers, the ring granularity and the per-stage detection probability, and together they map the sensitivity of the expected blast radius to each. Table III reports the granularity sweep at the reference detection probability, pairing each configuration's expected blast radius with its convergence cost and the marginal reduction it buys over the next-coarser configuration; Table IV reports the detection-probability sweep at the reference four-ring granularity. Every value is drawn from the frozen results file over the 25 evaluation seeds.

The two tables expose a sharp asymmetry in how the model responds to its levers. Along the granularity axis (Table III) the expected blast radius falls steeply at first and then flattens: the convergence cost climbs linearly from 0 to 5 while the marginal reduction collapses from 0.761 to 0.0353, so the last two rings buy almost nothing for two added soak periods. Along the detection axis (Table IV) the same four-ring structure spans an expected blast radius from 0.0156 to 0.5941, a range wider than the entire spread of the granularity sweep below the big-bang. The reduction relative to big-bang stays above 0.95 only while detection is strong and degrades to 0.4059 once the canary catches a fault only one time in five. Holding the partition fixed and varying only detection moves the outcome more than holding detection fixed and varying the partition, which locates the model's true sensitivity in observability rather than in ring count.

Adversarial reading. The detection-probability axis is also a threat surface. An adversary who can suppress the canary's ability to surface a fault, or, equivalently, a latent defect whose symptoms do not manifest during the soak window, drives the effective per-stage detection probability p down, and by Property (ii) and Eq. (10) this inflates the expected blast radius monotonically toward the big-bang value. A four-ring partition with a per-stage detection probability of 0.2 already exposes 0.5941 of the fleet in expectation, more than a four-ring partition is meant to contain, because a fault the gate does not detect advances through it exactly as it would through no gate at all. The implication is concrete: an attacker who corrupts an enforcement change has an incentive to make it quiet at the canary, and a defender's investment in detection quality per stage is therefore at least as load-bearing as the number of rings. Staging that is not paired with observability that actually fires during the soak is a partition without a gate, and the blast radius it bounds is no smaller than what an undetected fault reaches on its own.

VIII. Discussion

The results separate a benefit that is large and a cost that is fixed, and they locate where the benefit comes from. Four rings capture nearly all of the available blast-radius reduction. The move from a big-bang to a four-ring rollout reduces the expected blast radius by 0.9522, from 1.0 to 0.0478, and the further move to six rings shaves only another 0.0353 off the expected fraction of the fleet harmed. The marginal sixth-ring gain is tiny: it costs two more inter-ring stages of convergence latency, by Eq. (3), to lower the blast radius from 0.0478 to 0.0125. For most fleets the four-ring configuration sits at or near the interior operating point that H4 predicts, and the practical advice is to stage with a few small early rings rather than to keep subdividing.

Detection quality per stage matters as much as the number of rings. The detection-probability sweep shows that the same four-ring structure delivers an expected blast radius of 0.0156 when the canary catches a fault almost always ($p = 0.95$) but 0.5941 when it catches a fault only one time in five ($p = 0.2$). A finely partitioned fleet with a blind canary contains almost nothing, because a fault that the gate does not detect advances through it exactly as it would through no gate at all. Investment in observability, the monitoring, alerting, and health signals that let a soak period actually surface a fault [3], [10], is therefore at least as important as the ring count. An operator who has bought four rings but not the ability to detect a fault at the canary has paid the convergence cost without buying the containment.

The cost of containment is added convergence latency, and it is paid on every campaign. Because the convergence cost is the number of inter-ring stages and is incurred whether or not the change is faulty, a fleet that pushes many healthy changes pays the soak tax constantly while collecting the blast-radius benefit only on the small faulty fraction. This is the right way to read the tradeoff: staging is insurance whose premium is convergence latency on every release and whose payout is bounded damage on the rare bad one. Under remediation-timeline pressure [4], the convergence cost is not free latency but latency that must still clear a deadline, which is a further reason to prefer a few early rings over many. Three deployment rules follow directly. First, stage with small early rings, since a small f_1 bounds the blast radius of any fault caught at the canary. Second, invest first in canary observability, since detection probability governs containment as strongly as ring count. Third, stop adding rings once the marginal reduction no longer justifies the added convergence latency, which for the configurations studied here is at or before four rings.

IX. Threats to Validity

Construct validity. The model abstracts fault surfacing as a per-stage detection event at the gate boundary with a fixed probability. Real faults may surface with a delay rather than exactly at a soak boundary, may surface partially, or may depend on the workload exercised during the soak. A delayed-surfacing fault would let a change advance past the gate that should have caught it, raising the effective blast radius; this would shift absolute magnitudes but not the structural relationships, since Eq. (2) still governs the dependence on ring size and detection probability once p is reinterpreted as the probability of catching the fault before promotion.

Internal validity. Because the study is pre-registered and the evaluation seeds were inspected only once, the reported numbers are not the product of analytic search. The development of the model used separate seeds, and no ring fraction, fault rate, or detection probability was tuned to clear a threshold. The convergence cost is computed mechanically as the number of inter-ring stages and matches Eq. (3) exactly across all configurations.

External validity. The fault rate, the ring fractions, and the detection probabilities are documented priors, not measurements from any agency, so the absolute blast-radius magnitudes (0.0478 at the reference, 0.5941 at the lowest detection probability) should be read as illustrative of the model rather than as field estimates. A real canary population may not be representative of the fleet, so a fault that the canary cannot exhibit would not be caught at the gate regardless of p. The comparative findings, the large reduction from a few rings, the observability scaling, and the diminishing returns of finer staging, follow from the rollout accounting in Eqs. (2) and (3) rather than from the specific constants, and we expect them to transfer. Calibrating the fault rate, ring sizes, and detection probability against a real enforcement-deployment record is the natural next step and would convert the illustrative magnitudes into estimates.

Statistical validity. Intervals are BCa bootstrap intervals over 25 seeds, appropriate for the small-sample statistics reported here; we use interval non-overlap rather than significance testing throughout, avoiding the multiple-comparison pitfalls of repeated p-values.

X. Conclusion

A faulty enforcement script is dangerous because it runs everywhere, and a big-bang deployment lets it harm the whole fleet before anyone notices. Ring rollout bounds that harm by promoting a change through rings of increasing size and catching a fault at a gate, and we have quantified what it buys and what it costs. In a pre-registered simulation over 25 seeds, a four-ring rollout reduced the expected blast radius by 0.9522, from 1.0 to 0.0478, at a fixed convergence cost of three inter-ring stages; the expected blast radius rose from 0.0156 to 0.5941 as the per-stage detection probability fell from 0.95 to 0.2; and finer staging lowered the blast radius from 0.239 at two rings to 0.0125 at six but with marginal reductions of 0.761, 0.1912, and 0.0353 that shrink while the convergence cost grows linearly. Four rings capture nearly all of the available reduction, the marginal sixth-ring gain is tiny, detection quality per stage matters as much as the number of rings, and the cost is added convergence latency paid on every campaign. The deployment guidance is correspondingly concrete: stage with small early rings, invest first in canary observability, and stop adding rings once the marginal reduction no longer justifies the soak. Calibrating the model against a real enforcement-deployment record, and modelling delayed fault surfacing, is the next step toward turning these structural findings into field estimates.

References

[1] Microsoft, "Powershell desired state configuration (DSC)," https://learn.microsoft.com/powershell/dsc/overview, 2026, accessed 2026-06-20.

[2] Joint Task Force, "Security and privacy controls for information systems and organizations," National Institute of Standards and Technology, Tech. Rep. NIST Special Publication 800-53 Rev. 5, 2020.

[3] K. Dempsey, N. S. Chawla, A. Johnson, R. Johnston, A. C. Jones, A. Orebaugh, M. Scholl, and K. Stine, "Information security continuous monitoring (iscm) for federal information systems and organizations," National Institute of Standards and Technology, Tech. Rep. NIST Special Publication 800-137, 2011.

[4] Cybersecurity and Infrastructure Security Agency (CISA), "Binding operational directive 22-01: Reducing the significant risk of known exploited vulnerabilities," U.S. Department of Homeland Security, Tech. Rep., Nov. 2021, https://www.cisa.gov/news-events/directives/bod-22-01-reducing-significant-risk-known-exploited-vulnerabilities.

[5] B. Beyer, C. Jones, J. Petoff, and N. R. Murphy, Eds., *Site Reliability Engineering: How Google Runs Production Systems*. O'Reilly Media, 2016.

[6] G. Schermann, J. Cito, P. Leitner, U. Zdun, and H. C. Gall, "We're doing it live: A multi-method empirical study on continuous experimentation," *Information and Software Technology*, vol. 99, pp. 41-57, 2018.

[7] B. Efron, "Better bootstrap confidence intervals," *Journal of the American Statistical Association*, vol. 82, no. 397, pp. 171-185, 1987.

[8] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1994.

[9] C. Wohlin, P. Runeson, M. Höst, M. C. Ohlsson, B. Regnell, and A. Wesslén, *Experimentation in Software Engineering*. Springer, 2012.

[10] A. Basiri, N. Behnam, R. de Rooij, L. Hochstein, L. Kosewski, J. Reynolds, and C. Rosenthal, "Chaos engineering," *IEEE Software*, vol. 33, no. 3, pp. 35-41, 2016.

[11] J. O. Kephart and D. M. Chess, "The vision of autonomic computing," *IEEE Computer*, vol. 36, no. 1, pp. 41-50, 2003.

[12] National Institute of Standards and Technology, "Guide to enterprise patch management planning: Preventive maintenance for technology," National Institute of Standards and Technology, Tech. Rep. NIST Special Publication 800-40 Rev. 4, 2022.

[13] Verizon, "2026 data breach investigations report (DBIR)," Verizon Business, Tech. Rep., 2026, https://www.verizon.com/business/resources/reports/dbir/.

[14] H. Malla, "HygienePrio: Cyber-hygiene signal augmentation for EPSS-weighted vulnerability prioritization," https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper4, 2026, companion paper; source, code, and frozen results in the same repository.

[15] H. Malla, "Context-aware vulnerability prioritization for government endpoint fleets," https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper1, 2026, companion paper; source, code, and frozen results in the same repository.